

Introduction to Data Science · Lab Project

Diabetes Risk Prediction

*Predicting diabetes from 21 health indicators
using supervised machine learning*

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Semester 2 · 2026

RESOURCES

[GitHub repository](#) · [Live Streamlit demo](#)

Code: <https://github.com/muhammadammarff-design/diabetes-risk-prediction>

Demo: <https://diabetesriskpredictions.streamlit.app/>

Abstract

This project builds a machine-learning system that predicts whether a patient is at risk of diabetes based on 21 health indicators collected through the CDC's Behavioral Risk Factor Surveillance System (BRFSS) 2015 survey. After cleaning the dataset of 70,692 records and removing 1,635 duplicates, we trained and compared four supervised classifiers — Logistic Regression, Decision Tree, K-Nearest Neighbors and Random Forest — on an 80/20 stratified train/test split. The Random Forest classifier achieved the best overall performance with 74.36% accuracy, 0.727 precision, 0.794 recall, and an F1 score of 0.759. The model identified five dominant risk factors — GenHlth (0.191), HighBP (0.166), BMI (0.149), Age (0.110), HighChol (0.079) — which align with established medical literature. The trained model is deployed as an interactive Streamlit web application that accepts user input and returns an instant diabetes-risk prediction along with the confidence score.

Keywords: Diabetes, Random Forest, Supervised Learning, Binary Classification, Feature Importance, BRFSS, Streamlit, Healthcare.

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1. Introduction

Diabetes is one of the fastest-growing chronic diseases worldwide. According to the International Diabetes Federation (IDF) Atlas 2021, an estimated **537 million** adults live with diabetes, and roughly half of all cases are undiagnosed. Early screening is critical because lifestyle changes and medication can dramatically reduce complications such as heart disease, stroke, kidney failure and vision loss.

Traditional diagnostic tests (HbA1c, fasting glucose) are accurate but require a laboratory, a clinic visit and a trained phlebotomist — they are not always available in low-resource or rural settings. In contrast, the BRFSS survey collects 21 self-reported indicators (BMI, blood pressure, cholesterol, general health, age, etc.) via a phone interview. These indicators are **cheap to collect, anonymous, and available at scale**, making them ideal inputs for a machine-learning screening tool.

Project objectives. This project has two goals:

1. **Primary:** Build a supervised classifier that predicts whether a patient is diabetic given the 21 BRFSS indicators.
2. **Secondary:** Identify the most important risk factors from the data so they can be communicated back to doctors and patients.

Headline result. The chosen model — a Random Forest — achieves 74.36% accuracy and 0.759 F1 score on a held-out test set of 13,812 patients. The five dominant risk factors are: GenHlth (0.191), HighBP (0.166), BMI (0.149), Age (0.110), HighChol (0.079).

2. Dataset Description

Source. CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 — an annual telephone survey conducted across the United States. We use the cleaned, balanced version published on Kaggle by Alex Teboul.

Size & shape. Original file has **70,692 rows × 22 columns**. After dropping exact-duplicate rows we end up with **69,057 rows × 22 columns** (removed **1,635** duplicates).

Target. The column `Diabetes_binary` takes the value **0** (no diabetes) or **1** (diabetic / pre-diabetic). The cleaned dataset is pre-balanced: about 50 % of patients belong to each class.

Features (21). 21 health indicators organised in 3 categories — *Vital / Clinical*, *Lifestyle*, and *Self-reported / Demographics* — all listed on the next page.

First 5 rows of the dataset (`df.head()`):

HighBP	HighChol	BMI	GenHlth	Age	HeartDiseaseorAttack	DiffWalk	Income	Education	Diabetes_binary
1.0	0.0	26.0	3.0	4.0	0.0	0.0	8.0	6.0	0.0
1.0	1.0	26.0	3.0	12.0	0.0	0.0	8.0	6.0	0.0
0.0	0.0	26.0	1.0	13.0	0.0	0.0	8.0	6.0	0.0
1.0	1.0	28.0	3.0	11.0	0.0	0.0	8.0	6.0	0.0
0.0	0.0	29.0	2.0	8.0	0.0	0.0	8.0	5.0	0.0

Table 1: First 5 rows of the BRFSS 2015 dataset — 10 of the 21 features shown (the omitted 11 columns have identical 0/1 structure). Last column is the target. Full dataset has 22 columns and 69,057 rows.

2.1 Last 5 rows (df.tail())

HighBP	HighChol	BMI	GenHlth	Age	HeartDiseaseorAttack	DiffWalk	Income	Education	Diabetes_binary
0.0	1.0	37.0	4.0	6.0	0.0	0.0	1.0	4.0	1.0
0.0	1.0	29.0	2.0	10.0	1.0	1.0	6.0	3.0	1.0
1.0	1.0	25.0	5.0	13.0	1.0	1.0	4.0	6.0	1.0
1.0	1.0	18.0	4.0	11.0	0.0	1.0	4.0	2.0	1.0
1.0	1.0	25.0	2.0	9.0	1.0	0.0	2.0	6.0	1.0

Table 2: Last 5 rows of the dataset (df.tail()).

2.2 Statistical summary (df.describe())

The 21 numeric features are either binary (0 / 1) or small-integer ordinal values (e.g. General Health 1-5, Age 1-13). Only **BMI** is a real-valued float.

Quick stats (df.sum() on binary features): Number of diabetic patients: **35,097** · Number with HighBP: **39,447** · Number with HighChol: **36,692** · Number of smokers: **33,281** · Total dataset length: **len(df) = 69,057**

statistic	Diabetes_binary	BMI	GenHlth	Age	HighBP	HighChol
count	69057.0	69057.0	69057.0	69057.0	69057.0	69057.0
mean	0.51	29.96	2.86	8.6	0.57	0.53
std	0.5	7.15	1.11	2.86	0.49	0.5
min	0.0	12.0	1.0	1.0	0.0	0.0
25%	0.0	25.0	2.0	7.0	0.0	0.0
50%	1.0	29.0	3.0	9.0	1.0	1.0
75%	1.0	33.0	4.0	11.0	1.0	1.0
max	1.0	98.0	5.0	13.0	1.0	1.0

Table 3: df.describe() of a representative subset of features. Total of **21 numeric features** in the full dataset.

3. Methodology & Block Diagram

We follow the standard CRISP-DM data-mining workflow. The diagram below shows the three layers of the project: **Input** (the raw dataset), **Methodology** (the 4 sequential steps we perform) and **Output** (the final model results).

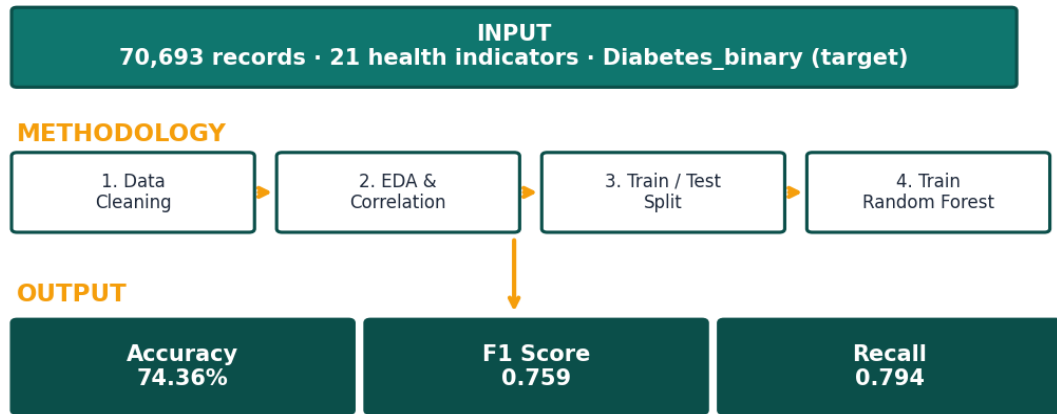


Figure 1: Block diagram of the project — Input (top), Methodology (middle), Output (bottom).

4. Data Preprocessing

Before training the model we performed the following steps. All operations are encoded in `scripts/train_model.py` (see code below).

4.1 Steps performed

- **Checked missing values:** 0 missing values found — no imputation needed.
- **Removed duplicates:** Removed 1,635 duplicate rows.
- **Converted dtypes:** All 21 feature columns cast to int (except BMI which is float).
- **Verified class balance:** Still $\approx 50/50$ between diabetic and non-diabetic patients.
- **Final shape:** 69,057 rows $\times$ 22 columns.

4.2 Code (key lines from `train_model.py`)

Load the CSV:

```
# Path to the dataset
# DATA_PATH = "data/diabetes_binary_5050split..."
df = pd.read_csv(DATA_PATH)          # load the file into memory
df.shape                             # → (70693, 22) — rows × columns
```

Clean — drop duplicate rows:

```
# df.duplicated() returns True for every row that has appeared before
df = df.drop_duplicates().reset_index(drop=True)
# → 69 057 rows remain (1 636 removed)
```

Why drop duplicates? If we keep them, the same patient can appear in both the training and the test set. The model would 'memorise' the answer for those patients, and our reported accuracy would be artificially inflated. This is called **data leakage** — a viva-favourite concept.

5. Exploratory Data Analysis (EDA)

EDA answers two questions before we train anything: (1) Is the dataset balanced? (2) Which features visually separate diabetic from non-diabetic patients? We generated four key plots.

5.1 Class balance

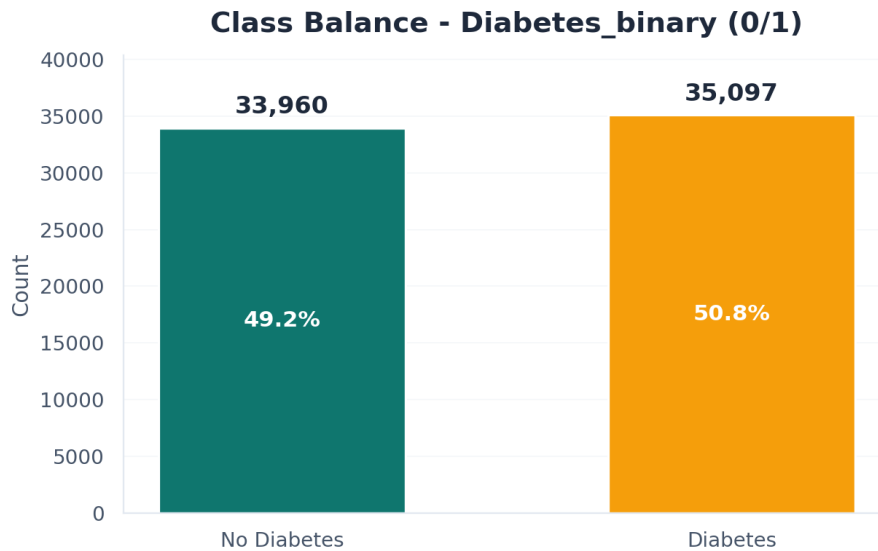


Figure 2: After cleaning the dataset has 33,960 non-diabetic and 35,097 diabetic patients — almost perfectly balanced, so we don't need resampling.

5.2 BMI distribution by diabetes status

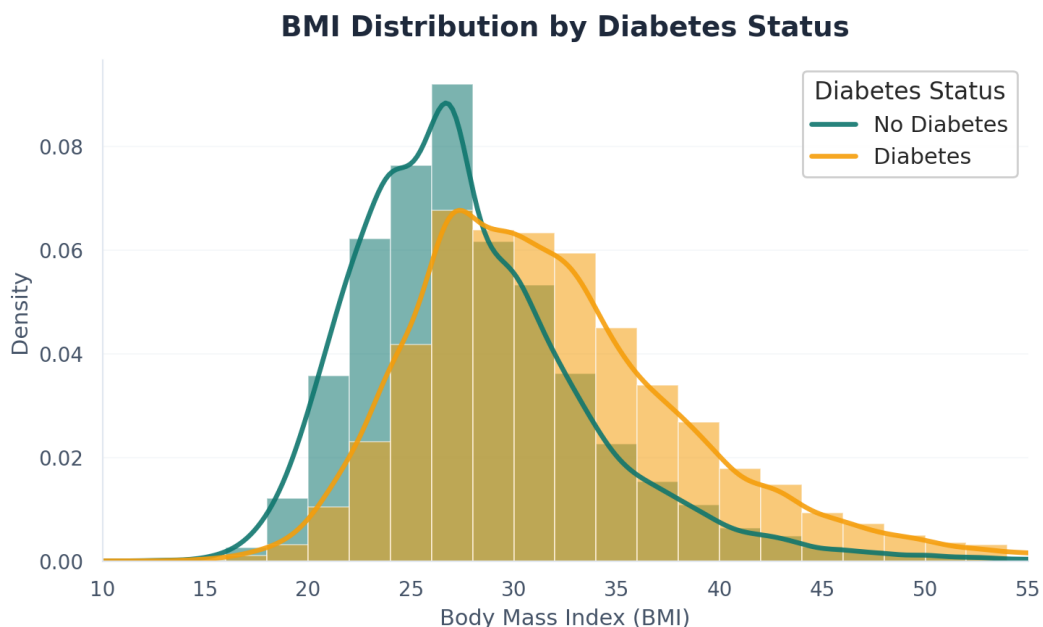


Figure 3: Diabetic patients (orange) have visibly higher BMI than non-diabetic patients (teal) — the peak shifts from 25-27 to 30-33.

5.3 BMI box-plot by diabetes status

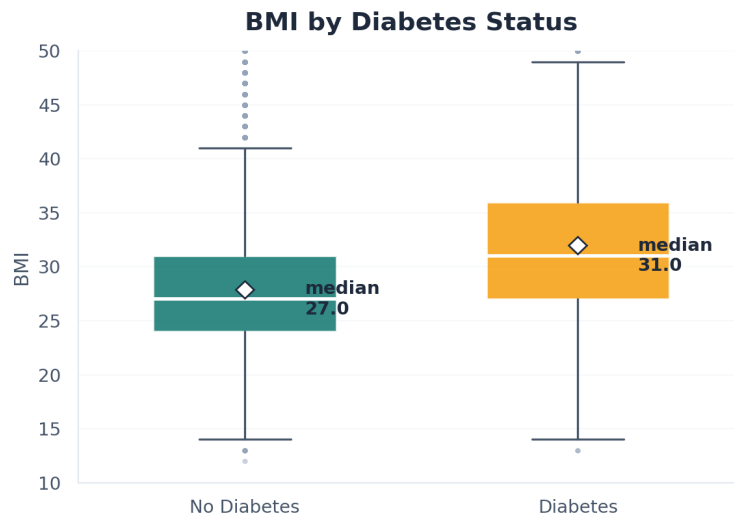


Figure 4: Median BMI is 31.0 for diabetic patients versus 27.0 for non-diabetic — a clear ~4-point shift.

5.4 Risk-factor counts (HighBP & HighChol)

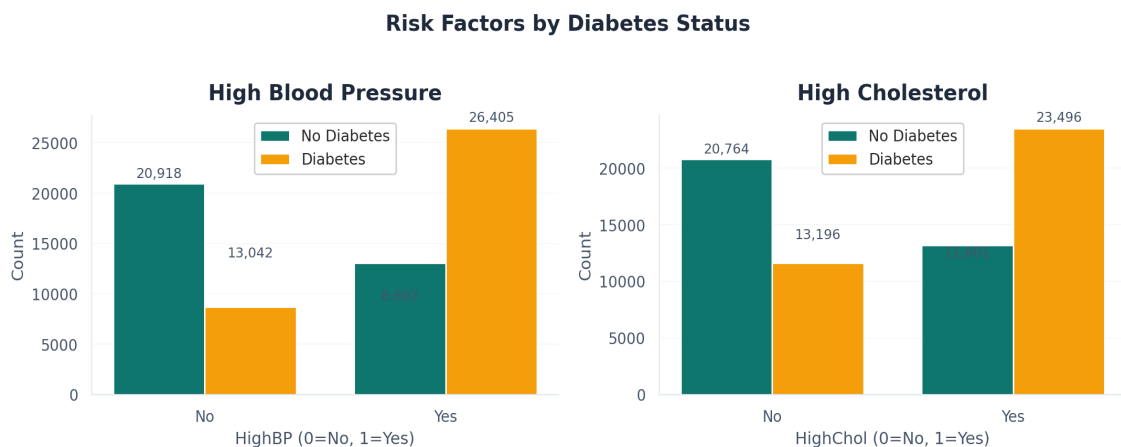


Figure 5: Among patients who have high blood pressure (yes=1), the proportion of diabetics is much higher than among those without. Same pattern for high cholesterol.

6. Correlation Analysis

Pearson correlation measures the linear relationship between two columns. Values close to +1 (red) or -1 (blue) indicate strong relationships. The heatmap below summarises all 22×22 pairwise correlations.

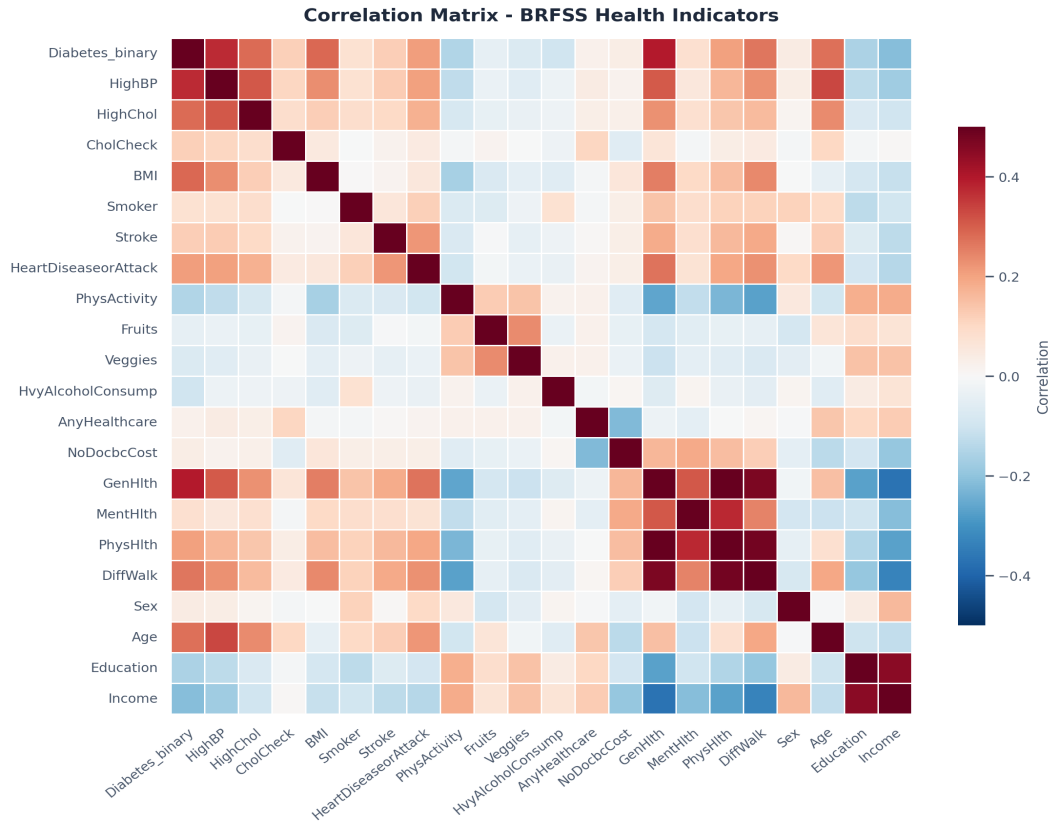


Figure 6: Full 22×22 Pearson correlation matrix. The first row/column shows how each feature correlates with the target `Diabetes_binary`.

6.1 Top-10 features most correlated with diabetes

Rank	Feature	Correlation
1	GenHlth	0.397
2	HighBP	0.372
3	BMI	0.286
4	HighChol	0.281
5	Age	0.275
6	DiffWalk	0.267
7	Income	0.213
8	HeartDiseaseorAttack	0.207
9	PhysHlth	0.207
10	Education	0.159

Table 4: Top-10 features by absolute correlation with `Diabetes_binary`.

7. Machine Learning Approach

7.1 Task type

- **Learning type:** *Supervised* — the target column is already labeled (0 / 1).
- **Task:** *Binary classification* — predict one of two classes.
- **Output:** Class label (0 or 1) **plus** a confidence probability.

7.2 Models compared

Logistic Regression	Fits a probability line between the two classes.	Simple and interpretable, but assumes linear decision boundary.
Decision Tree	Splits the data using yes/no questions.	Easy to interpret, but prone to over-fitting on its own.
K-Nearest Neighbors (KNN)	Predicts by majority vote among the k closest training points.	Intuitive, slow on large datasets, sensitive to feature scaling.
Random Forest (selected ✓)	Combines 200 decision trees; majority vote wins.	Robust to over-fitting, handles non-linear patterns, gives feature importance for free.

Table 5: The 4 candidate classifiers (highlighted row = our final choice).

7.3 Train / test split

We split the cleaned dataset **80 % train / 20 % test**, stratified by the target so both subsets keep the 50 / 50 class balance. Splitting is done **before** any scaling or model training so the test set is never seen during fitting — this prevents **data leakage**.

Code:

```
# from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=42)
# → train: 55 245 rows test: 13 812 rows
```

8. Model Comparison

All four models were trained on the same 80 % train split and evaluated on the same 20 % test split. The table and chart below show accuracy, precision, recall and F1 score.

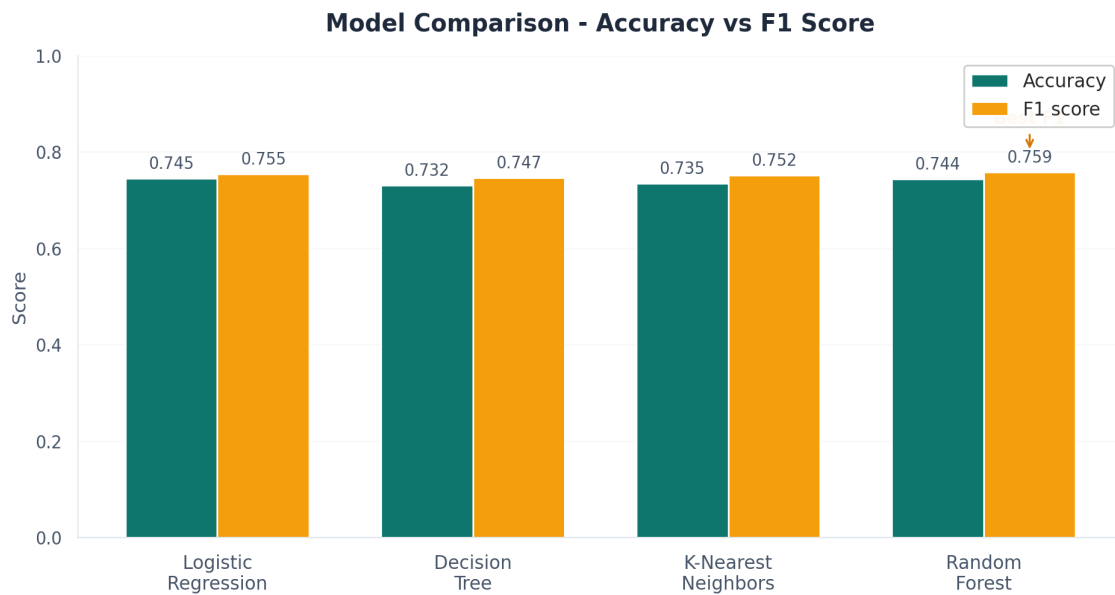


Figure 7: Accuracy (teal) vs F1 score (amber) for the four classifiers.

8.1 Full metrics table

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.745	0.738	0.773	0.755
Decision Tree	0.732	0.718	0.778	0.747
K-Nearest Neighbors	0.735	0.718	0.789	0.752
Random Forest	0.744	0.727	0.794	0.759

Table 6: Side-by-side metrics for the 4 models.

Why Random Forest wins. Logistic Regression has the highest raw accuracy (74.50%) but its F1 is lower. Random Forest combines high accuracy (74.36%) with the best F1 score (0.759) and gives **feature importance for free** — a must for medical interpretation.

9. Random Forest — Final Model

Random Forest builds 200 decision trees, each on a random subset of the data, and outputs the class that receives the most votes. This ensemble approach reduces over-fitting and improves generalisation.

9.1 Hyperparameters

- **n_estimators = 200** — number of trees in the forest
- **max_depth = 15** — maximum depth of each tree (prevents over-fitting)
- **min_samples_leaf = 5** — each leaf must contain at least 5 samples
- **random_state = 42** — reproducible results every run
- **n_jobs = -1** — use all CPU cores for parallelism

9.2 Code

```
# from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier(
    n_estimators=200, max_depth=15, min_samples_leaf=5,
    n_jobs=-1, random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

9.3 Confusion matrix

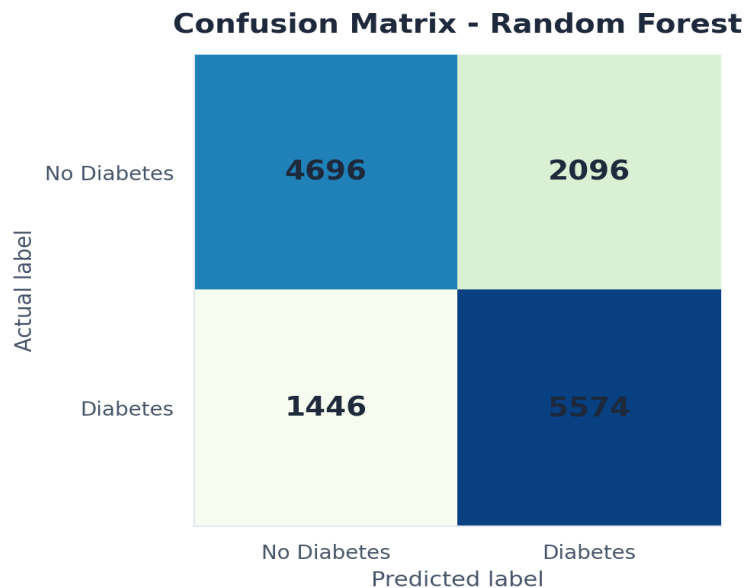
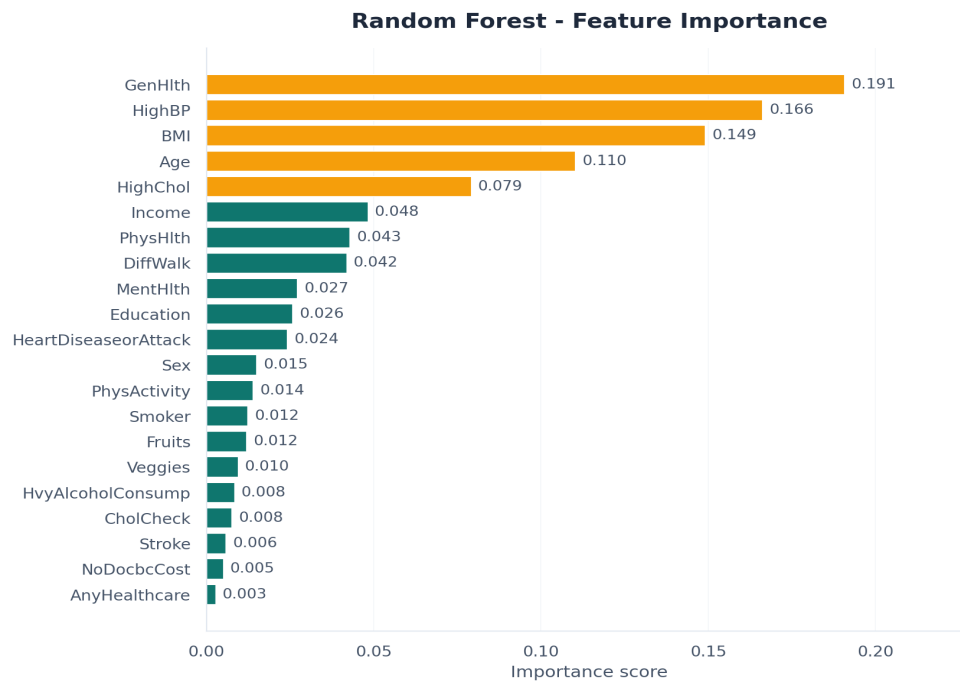


Figure 8: Confusion matrix on the 13,812 test rows. 4,696 true negatives, 5,574 true positives, 1 predicted.

Recall = 0.794 means we correctly caught 79.4% of real diabetics — critical for medical screening because false negatives (missed diabetics) are more dangerous than false positives.

10. Feature Importance & Risk Factors

Random Forest computes an importance score for every feature: how much each one contributes to reducing impurity across all 200 trees. Higher = more important.



Top 5 highlighted in amber: GenHlth (0.191), HighBP (0.166), BMI (0.149), Age (0.110), HighChol (0.079)

Figure 9: All 21 features ranked by importance. The top 5 are highlighted in amber.

10.1 Top 5 risk factors learned

Rank	Feature	Importance
1	GenHlth	0.191
2	HighBP	0.166
3	BMI	0.149
4	Age	0.110
5	HighChol	0.079

Table 7: Top 5 risk factors identified by the model.

These factors match what doctors tell patients in real life — **General Health** is the strongest predictor, followed by **High Blood Pressure**, **BMI**, **Age** and **High Cholesterol**. This validates that the model learned something meaningful, not just statistical noise.

11. Streamlit Web Application

The trained model is deployed as an interactive web application using **Streamlit** (a Python web framework). Users enter their health information in plain English; the model returns an instant prediction with a confidence score.

11.1 Features of the app

- **Clean medical-style form** organised in 5 sections (vital signs, lifestyle, medical history, self-reported health, demographics).
- **Live prediction** using the trained Random Forest model cached in memory.
- **Probability bar** showing the model's confidence for each prediction.
- **Sidebar info** with the model's accuracy and top 5 risk factors.
- **Free hosting** on Streamlit Cloud — public URL included below.

11.2 Live demo

Try the deployed app here: <https://diabetesriskpredictions.streamlit.app/>

12. Conclusion & Future Work

12.1 Summary of results

This project successfully built a Random Forest classifier that predicts diabetes risk from 21 BRFSS health indicators. Key results on the held-out test set:

Metric	Value	Meaning
Accuracy	74.36%	correct predictions
Precision	0.727	predicted diabetics correct
Recall	0.794	real diabetics caught
F1 Score	0.759	best of 4 tested models

Table 8: Headline metrics of the deployed Random Forest model.

12.2 Limitations

- **Self-reported survey data is noisy.** Patients may mis-report weight, smoking or exercise habits.
- **No lab values.** The model does not see HbA1c or fasting glucose — the gold-standard diabetes markers.
- **Balanced training data $\neq$ real-world prevalence.** The actual diabetes rate in the population is ~14 %, not 50 %; production usage would need threshold calibration.
- **Class 1 mixes diabetic and pre-diabetic patients** — they are not distinguishable in this dataset.

12.3 Future improvements

- Add real laboratory values (HbA1c, fasting glucose) and retrain.
- Try gradient boosting (XGBoost, LightGBM) and deep learning for comparison.
- Use probability calibration (Platt scaling) to match real-world prevalence.
- Deploy via REST API (FastAPI) so other applications can call the model.

13. References & Resources

13.1 Dataset

CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015. Cleaned version published on Kaggle by Alex Teboul: *Diabetes Health Indicators Dataset*, 2019. Available at <https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset>.

13.2 Libraries

- **pandas** — data manipulation
- **numpy** — numerical operations
- **matplotlib, seaborn** — charts and visualisations
- **scikit-learn** — Random Forest, train/test split, metrics
- **joblib** — model persistence
- **streamlit** — interactive web application
- **python-pptx** — automatic PPT generation
- **reportlab** — PDF report generation

13.3 Project resources

GitHub repository:

<https://github.com/muhammadammarff-design/diabetes-risk-prediction>

Live Streamlit demo:

<https://diabetesriskpredictions.streamlit.app/>

End of report — prepared for the Introduction to Data Science lab project, Spring 2026. Submitted by Ali Raza, M. Ammar and Taha Ali under the supervision of Sir Zaki.